

Solved Paper	Maharashtra HSC Exam. March 2016	Biology
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Time : 3 Hours

Maximum Marks : 70

General Instructions:

1. All questions are compulsory.
2. Draw neat and labelled diagram wherever necessary.
3. Figures to the right indicate full marks.
4. Answers to the question in Section –I and Section –II should be written in two separate Answer books.
5. Questions form Section –I attempted in the Answer book of Section –II and vice –versa will be assessed/not be given any credit.
6. Answer to every new question must be started on a new page.

SECTION - I (BOTANY)

1. Select and write the most appropriate Answers form the given alternatives for each sub-question: [7]

- (i) The phenotypic ratio of incomplete dominance is _____
- (a) 2:1 (b) 1:2:1
- (c) 1:1:1 (d) 1:1:2

Answer (b) 1:2:1

- (ii) The number of purines in a segment of DNA molecule is 68. What will be the number of pyrimidines in this segment?
- (a) 34 (b) 43
- (c) 68 (d) 86

Answer (c) 68

- (iii) Alcoholic fermentation is brought about by -----
- (a) Lactobacillus (b) Saccharomyces
- (c) Trichoderma (d) Streptomyces

Answer (b) Saccharomyces

- (iv) Which of the following is not a photosynthetic pigment?
 (a) Carotene (b) Xanthophyll
 (c) Phycobillins (d) Anthocyanin

Answer (d) Anthocyanin

- (v) Which of the following is a stop codon?
 (a) UAG (b) UAC
 (c) AUG (d) UCA

Answer (a) UAG

- (vi) Pyruvate undergoes oxidative decarboxylation to produce _____
 (a) 2-PGA (b) α -Ketoglutarate
 (c) Succinyl-Co-A (d) Acetyl-Co-A

Answer (d) Acetyl-Co-A

- (vii) Which day is observed as 'World Environment Day'?
 (a) 21st May (b) 5th June
 (c) 25th September (d) 13th December

Answer (b) 5th June

2. (A) Answer in 'one' sentence only

[6]

- (i) **What is test cross?**

Answer: Test Cross: A genetic cross between individual with unknown genotype for a particular trait with a recessive plant for that trait.

- (ii) **What is mycoheribicide?**

Answer: Mycoheribicide: Microbes which are used as herbicides are called mycoherbicide. For example – Phytophthora palmivora

(iii) **What is Anticodon?**

Answer: Anticodon: It is a sequence of three nucleotides present on the anticodon loop of transfer RNA corresponding to a complementary codon in messenger RNA.

(iv) **What is Humification?**

Answer: Humification: It is defined as the formation of humus which is a dark coloured amorphous substance. It is a reservoir of nutrients.

(v) **How CO₂ makes idlies puffy?**

Answer: The bubbles of CO₂ trapped in gluten makes idlies puffy.

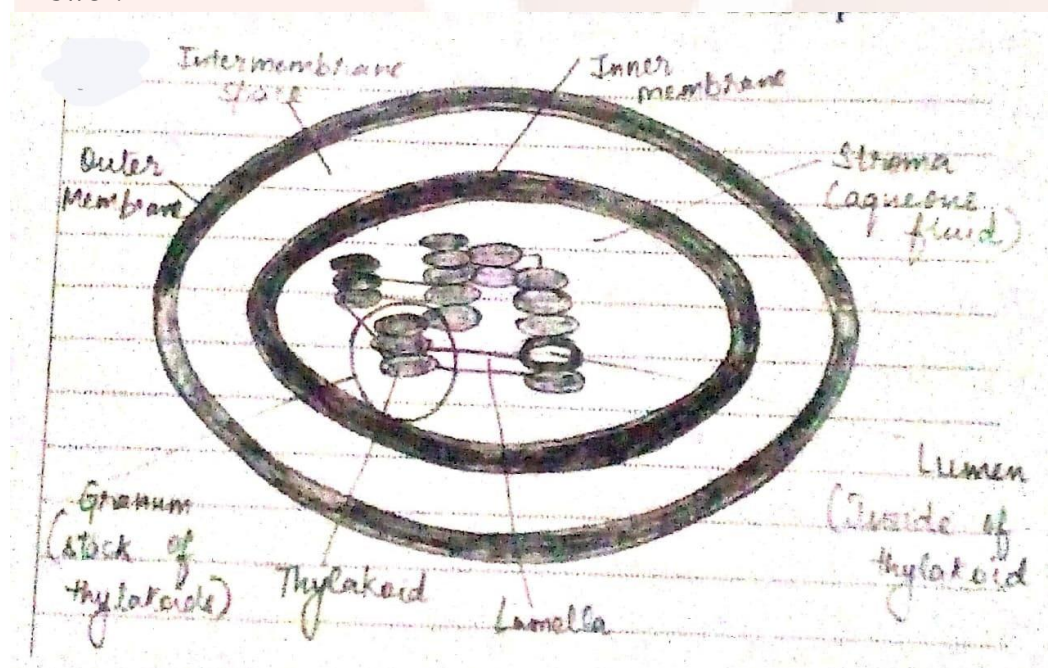
(vi) **What is ecological succession?**

Answer: Ecological succession: The process of change in species composition of a given area overtime is called ecological succession.

(B) Sketch and label the 'Ultra structure of Chloroplast'.

[2]

Answer:



(C) Answer the following (Any Two):

[4]

(i) Write a short note on 'Mutational breeding'

Answer: Mutational Breeding: It is a breeding by mutation through the use of chemicals or radiations and utilized in developing desirable traits in an organisms.

(ii) Enlist the advantages of Biogas.

Answer: Advantage of Biogas: The advantages of biogas are –

- a. It is a renewable source of energy,
- b. It is cheap and safe to use
- c. It reduces greenhouse effect
- d. It is used for cooking, domestic lighting as well as small scale industries.

(iii) Explain 'Carbon Cycle'.

Answer: The carbon cycle is being explained below:

- a) Carbon cycle is the cyclic flow of carbon in the atmosphere.
- b) The major processes which help in circulation of carbon in the atmosphere and through the organism is photosynthesis and respiration.
- c) Respiration requires carbohydrates and oxygen and releases carbon dioxide, water and energy. Photosynthesis requires carbon dioxide and water and produces carbohydrates and oxygen.
- d) The carbon trapped in organisms which is not released by respiration is released during decomposition in the atmosphere or gets converted to fossil fuels.
- e) The carbon trapped in fossil fuels is released during combustion.

(iv) Give the floral adaptations of chiropterophily.

Answer: The floral adaptations of chiropterophily are:

- a) The flowers are large and stout to allow bats to hold on to them.
- b) They open during the night.
- c) They give out a rotten fruit like fermenting fruity odour.
- d) They produce copious nectar.
- e) They produce a large number of stamens

3. (A) Answer the following (Any Two)

[6]

(i) Why the ratio in pleiotropy is 2:1? Explain it with example.

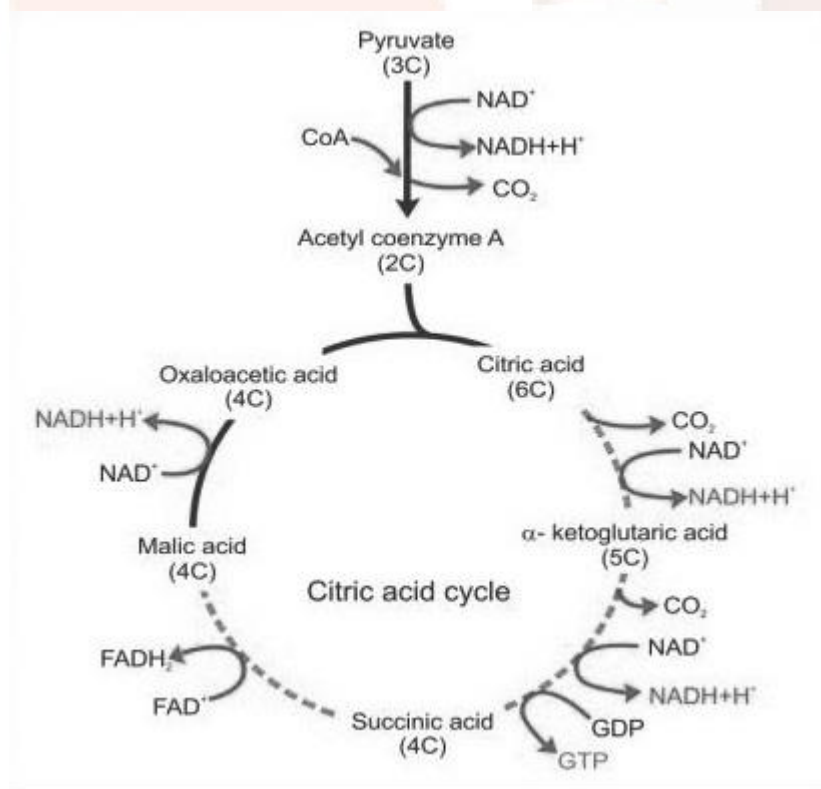
Answer: Pleiotropy: When a single gene controls two or more characteristics (or traits), it is called pleiotropy. The gene is called pleiotropic gene. As per Mendel's principle, one gene controls one character but sometimes single gene produces two related or unrelated phenotype expressions.

For example: The disease sickle cell anaemia is caused by this phenomenon. The disease is caused by gene Hb^s . In a normal or healthy individual, the gene is Hb^A , which is dominant. A heterozygote is a carrier and shows mild signs of anaemia and the RBC's become sickle shaped with a deficiency of oxygen. The homozygous recessive individuals however show fatal anaemia and die. Hence, if two carriers for sickle cell anaemia mate, the children are formed in the ratio of 1:2:1 with one normal, two carriers and 1 diseased. Because the diseased child dies, the observed ratio of the surviving children is 1:2.

Thus, pleiotropy shows a 2:1 ratio and not the normal Mendelian 3:1 ratio.

(ii) Give the schematic representation TCA cycle.

Answer:



(iii) Differentiate between cyclic and non-cyclic photophosphorylation

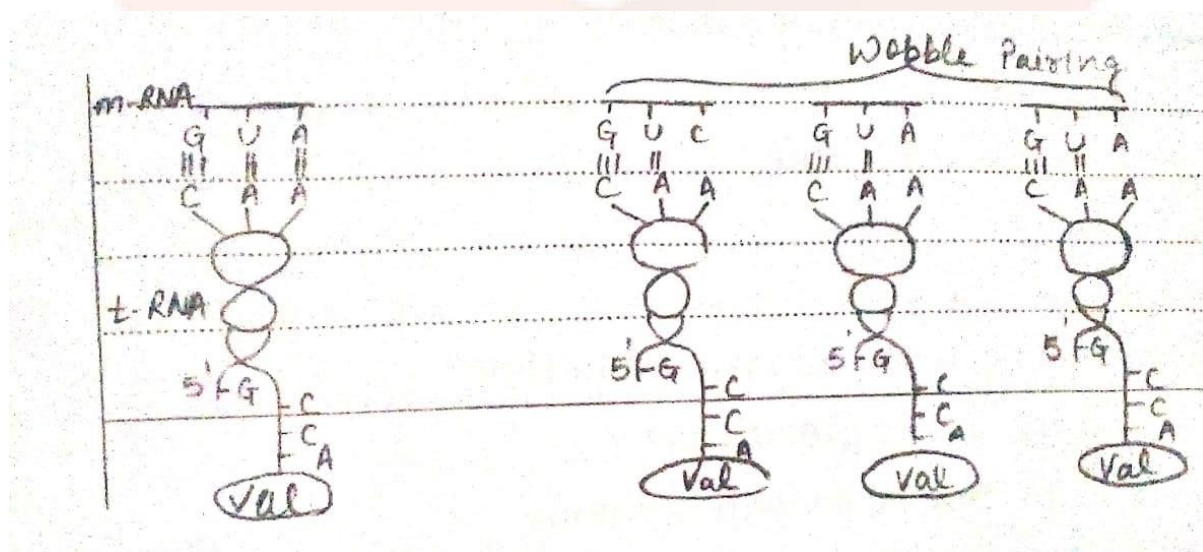
Answer:

S.No.	Cyclic photophosphorylation	Non-cyclic photophosphorylation
1	It is related with PS I	It is related with both PS I and PS II
2	Photolysis of water and evolution of oxygen do not occur.	Photolysis of water and evolution of oxygen occurs
3	Photophosphorylation occurs at two places	Photophosphorylation takes place at one place
4	NADP is not reduced	NADP is reduced to NADPH ₂

(B) Give diagrammatic representation to shows a perfect pairing and any 'two' Wobble pairing. [3]

Answer. The genetic code is degenerate, as 61 codons are available for 20 amino acids. Two or more codons can specify the same amino acid, e.g. GGG, GGGA, GGC and GGU code for glycine. This was explained by wobble hypothesis.

This hypothesis explains that in codon anticodon pairing the third base may not be complementary. The third base of the codon is called wobble base and this position is called wobble position. The actual base pairing occurs at first two positions only for example- GUU, GUC, GUA and GUG code for amino acid valine.



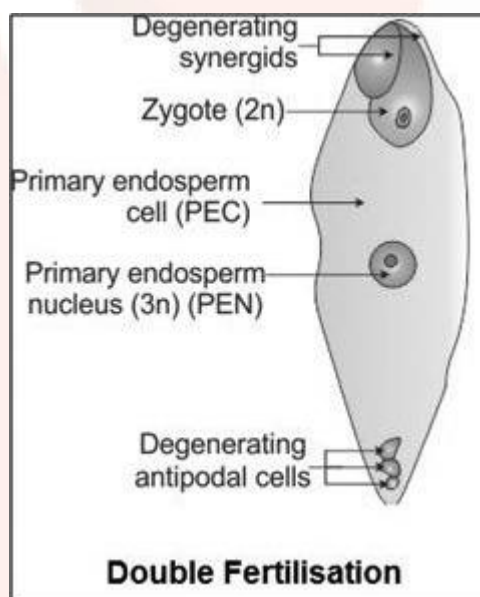
4) What is double fertilisation? Describe the process in brief. [7]

Or

Define r-DNA technology .Give the basic steps in r-DNA technology and give aye ‘three ‘examples of the therapeutic products produced by r-DNA technology.

Answer.

Double fertilization is the process of fusion of one male gamete with the egg along with the union of the second male gamete with the two polar nuclei or the secondary nucleus. Out of the two male gametes, one fuses with the egg to carry out generative fertilization or syngamy. It gives rise to a diploid zygote or oosphere. The nucleus of the second male gamete fuses with two haploid polar nuclei or diploid secondary nucleus of the central cell to form a triploid primary endosperm nucleus called PEN. The central cell is called primary endosperm cell, PEC. This is called the vegetative fertilization or triple fusion.



OR

r-DNA technology: It is the technique of manipulating the genome of a cell or organism to change the phenotype desirably .

The basic steps involved in the process are:

1. Isolating genomic DNA or a donor .The cell or organism from which the required gene is taken is called donor.
2. Fragmenting this DNA using molecular scissors (enzymes Restriction Endonuclease)
3. Screening the fragments for a desired gene.

4. Inserting the fragments with desired gene into a cloning vector to develop a recombinant DNA.
5. Introducing the recombinant vector into a competent host cell.
6. Culturing these cells to obtain multiple copies or clones of desired fragment of DNA.
7. Using these copies to transform suitable host cells so as to express the desired gene.

Examples of therapeutic products produced by r-DNA technology are –

1. Vaccines : Hepatitis B
2. Human Hormones : Insulin
3. Blood proteins : Erythropoietin

SECTION –II (ZOOLOGY)

Q.5) Select and write the most appropriate Answer from the given alternatives for each sub –question:

(i) If centromere is situated in the middle of the chromosome, it is called_____.

- | | |
|--------------------|-----------------|
| (a) Metacentric | (b) Acrocentric |
| (C) Submetacentric | (d) Telocentric |

Answer: (a) Metacentric

(ii) Which of the following is useful in treatment of burns and wound healing?

- | | |
|----------------------------------|---------------------------|
| (a) Tissue plasminogen activator | (b) Tissue growth factor |
| (C) DNase | (d) Bovine growth hormone |

Answer: (b) Tissue growth factor

(iii) Erythroblastosis foetalis is caused when mother is_____.

- | | |
|-----------------------|-----------------------|
| (a) Rh ^{+ve} | (b) With antibody 'A' |
| (C) Rh ^{-ve} | (d) With antibody 'B' |

Answer: (C) Rh^{-ve}

(iv) Deposition of fatty substances in the lining of arteries results in _____.

- | | |
|----------------------|---------------------|
| (a) Arteriosclerosis | (b) Atherosclerosis |
| (c) Hyperglycemia | (d) Hypotension |

Answer: (b) Atherosclerosis

(v) Struggle between cow and cow for getting grass is called _____.

- | | |
|---|-----------------------------|
| (a) Inter-specific struggle | (b) Environmental struggle |
| (c) Struggle against natural calamities | (d) Inter-specific struggle |

Answer: (d) Inter-specific struggle

(vi) In which type of adaptation, forelimbs are modified into wings?

- | | |
|--------------------------|---------------------------|
| (a) Aquatic adaptations | (b) Volant adaptations |
| (C) Arboreal adaptations | (d) Cursorial adaptations |

Answer: (b) Volant adaptations

(vii) Mating of two closely related individuals within the same breed is called _____.

- | | |
|-------------------|---------------------|
| (a) In-breeding | (b) Out -breeding |
| (C) Out –crossing | (d) Cross -breeding |

Answer: (a) In-breeding

6 (A) Answer the following in 'one' sentence each: [6]

(i) Visit of veterinary doctor to dairy farm is mandatory. Give reasons.

Answer: Visit of a veterinary doctor to dairy farm is mandatory as it will ensure a proper and disease free breed, suitable and safe farm conditions, proper feed and health care. Identification of health problems, diseases and rectification can be done on time.

(ii) Why aquatic animals can afford to be ammonotelic?

Answer: For elimination of ammonia, large quantity of water is required. About 300 to 500 ml of water is required. Due to this reason, ammonotelism is seen in aquatic invertebrates, bony fishes, tadpole larva of frog.

(iii) Why PUC is mandatory for vehicles?

Answer Pollution under control (PUC) certificate is mandatory for all vehicles in order to keep a check on the amount of pollutants being emitted by these sources.

(iv) Define organic evolution.

Answer. Organic Evolution is defined as a slow, gradual, continuous and irreversible changes through which present day complex form have descended from their simple pre-existing forms of the past.

(v) Give the genotype of Turner's syndrome.

Answer. The genotype of Turner's syndrome is $(44+X_0)$.

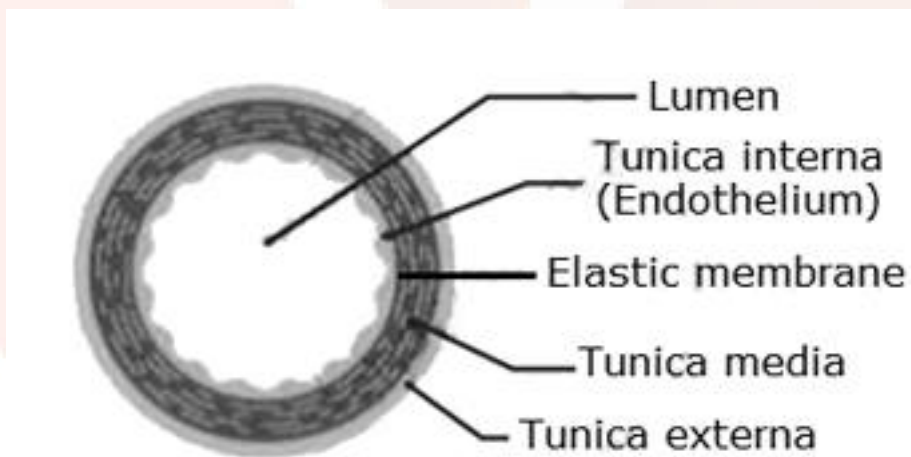
(vi) Write down the full form of R.F.L.P. is Restriction

Answer. Full form of R.F.L.P. is Restriction fragment length polymorphism.

(B) Sketch and label T.S of vein.

[2]

Answer:



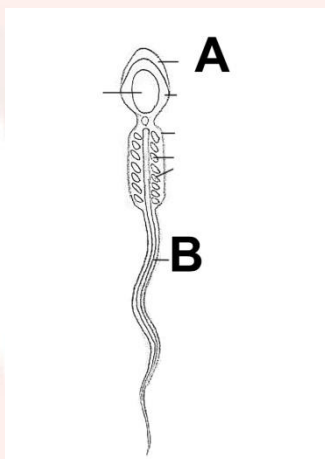
(C) Attempt any two of the following: [4]

(i) Give any 'two' factors responsible for genetic variation.

Answer: Two factors responsible for genetic variation are;-

1. Gene flow: Transfer of gene between populations that differ genetically from one another is called gene flow.
2. Gene Mutation: Changes in the chemical makeup of a gene is called gene mutations or point mutations.

(ii) Give the name and function for 'A' and 'B' from the diagram given below:



Answer:

A- Acrosome

B-Tail

(iii) Write a note on 'artificial acquired active immunity'.

Answer: It is acquired artificially by vaccination. Vaccines contain dead or live but attenuated pathogens or toxoids. These consist of microbial components or toxins secreted by the pathogens. The vaccine is introduced into the body to stimulate the production of antibodies by the immune system. Examples: Polio vaccine, BCG vaccine.

(iv) Give the economic importance of 'fisheries'.

Answer: The economic importance of fisheries is:

- (a) It is source of nutritious food
- (b) It shows commercial value like fish oil, fish meal and fertilization.
- (c) By products are widely used in paints, soaps, oils, and medicines.
- (d) Prawns and lobsters have export market value.
- (e) It provides job opportunity and self-employment.
- (f) It improves productivity and economy of the nation.

7. (A) Attempt any two of the following: [6]

(i) Explain sex determination in human beings.

Answer: Sex Determination in human beings.

The chromosomal mechanism of sex determination is of XX-XY type. In human beings, the nucleus of each cell contains 46 chromosomes or 23 pairs of chromosome and 1 pair is of sex chromosome responsible for determination of sex.

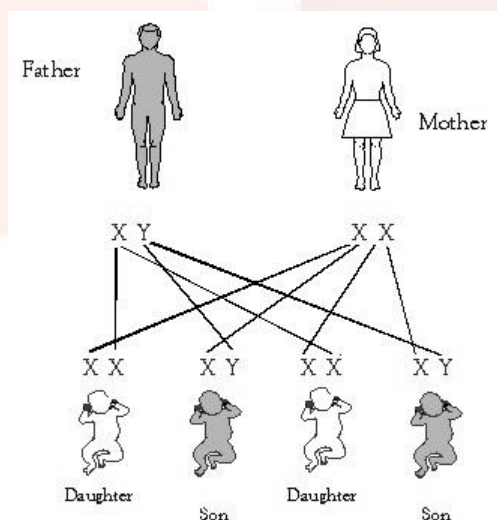
In female, two homomorphic sex chromosomes are XX.

In male, two heteromorphic sex chromosomes are XY.

The genotypes of male and female is:

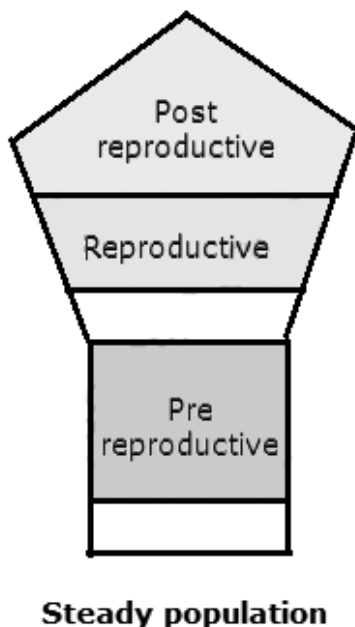
Male: 46 chromosomes = 44 Autosomes + XY sex chromosomes

Female: 46 chromosomes = 44 Autosomes + XX sex chromosomes



(ii) Describe steady population with the help of a pyramid

Answer: Steady Population- When the pre-reproductive and post-reproductive age group is same then the population remains steady.



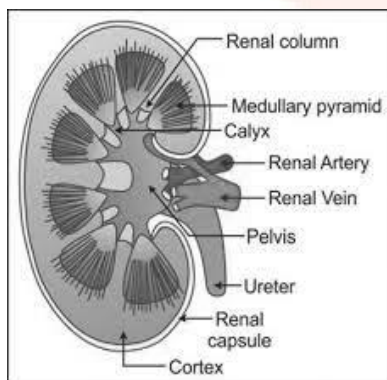
(iii) Illustrate any 'three' genes which can be used in gene therapy.

Answer: The three genes that can be used in gene therapy:

- (a) Tissue growth factor – beta (TGF- β): It promotes new blood vessels and epidermal growth. It is useful in wound healing and treatment of burns.
- (b) Tissue plasminogen activator (TPA): It is used to prevent or reverse blood clots.
- (c) Human blood clotting factor VIII: It is used to aid in the clotting of blood and to treat patients suffering from haemophilia

(B) Sketch and label 'L.S. of human kidney' [3]

Answer:



8. With the help of diagrammatic representation, explain the process of gametogenesis. [7]

OR

Describe the structure of cerebrum. Add a note on its function. [7]

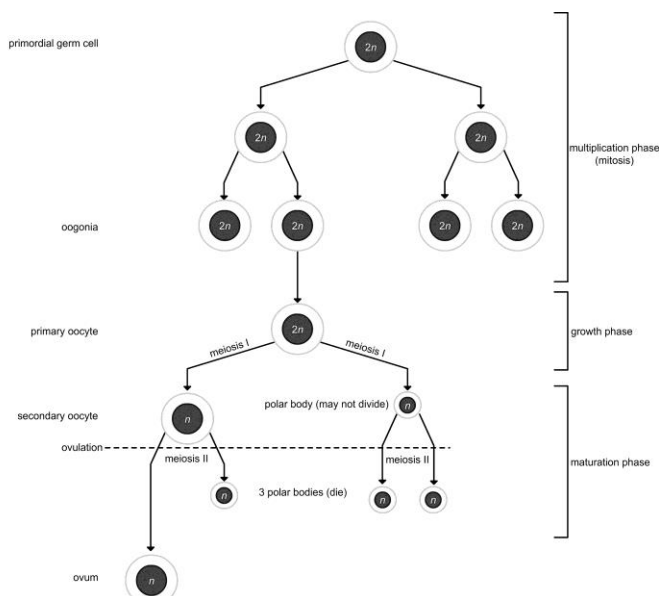
Answer. Gametogenesis- It is the process of formation of gametes in sexually reproducing animals. These animals show presence of somatic cells and germinal cells. Somatic cells form organ of the body and multiply mitotically.

The germinal cells form the gamete cells by meiosis and mitosis.

The process of formation of female gamete i.e. ovum is called oogenesis.

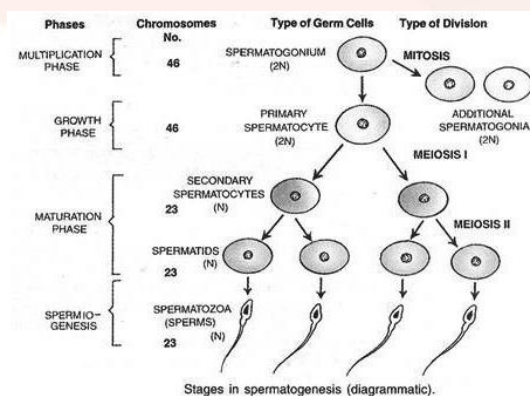
The process of formation of male gamete i.e. Spermatozoa is called spermatogenesis.

1. Oogenesis: - The process of oogenesis occurs in female gonad called ovary. It has three phases:
 - (i) Multiplication phase: Germinal cells undergo mitosis to form large number of oogenia. Oogenia in human being are formed in ovary of female baby even before her birth.
 - (ii) Phase of Growth: Just before puberty, under the influence of follicle stimulating hormone, one of the oogenium grows in size. Growth in size of oogenium is larger than seen in spermatogenesis. This grown up into a cell is called Primary Oocyte.
 - (iii) Maturation Phase: Primary oocyte undergoes maturation on meiotic division.
 - Meiotic I division of primary oocyte shows equal nuclear division but unequal cytoplasmic division.
 - Secondary oocyte and polar bodies undergo II meiotic division.



2. Spermatogenesis: The process of spermatogenesis takes place in male gonads called testis. Each testis has seminiferous tubules which are lined by cuboidal epithelium called germinal epithelium. The cells of germinal epithelium undergo spermatogenesis to produce sperms. In between germinal cells sertoli cells are present, which provide nourishment to the sperms. Germinal cells in testis are known as primary germinal cells. Primordial cells pass through three phases-

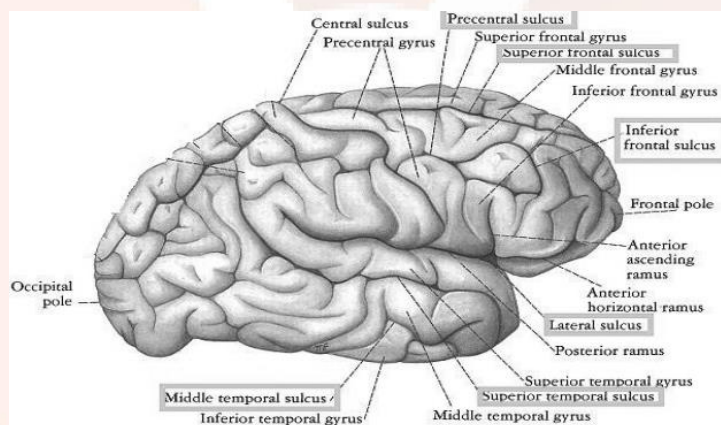
- (i) **Multiplication phase:** Primordial cells undergo repeated mitotic divisions to produce large number of spermatogonia. Each spermatogonium is diploid (2n).
- (ii) **The Growth Phase:** Spermatogonium cell accumulates food and grows in size and is called primary spermatocyte.
- (iii) **Maturation Phase:** The Primary spermatocyte undergoes first meiotic or maturation division. The homologous chromosomes start pairing. Each homologous chromosome splits longitudinally.



OR

Cerebrum is the largest portion of the brain and constitutes about 80-85 percent of the weight of the brain. It is divided into two cerebral hemispheres by a median longitudinal fissure. A cavity filled with cerebrospinal fluid (CSF) is present in each hemisphere. It is called the lateral ventricle. The outer surface of the cerebral hemisphere is called the cerebral cortex. The cortex contains cell bodies of the neuron and is greyish hence, it is called grey matter. The grey matter has many folds (that is gyri) and grooves (that is sulci). They increase the surface area to accommodate more nerve cells. A higher number of convolutions lead to greater intelligence. The inner surface of the cerebral hemisphere is called the cerebral medulla. The medulla consists of axons of nerve fibres and is called white matter. Each cerebral hemisphere is divided into four lobes- frontal lobe, parietal lobe, occipital lobe and temporal lobe. The frontal and Occipital lobes are separated by a central sulcus the parietal and temporal lobes are separated by a lateral sulcus, while the parietal and occipital lobes are separated by a parieto-optical sulcus.

The cerebrum determines intelligence in animals. It is also the centre of learning and memorizing in the brain.



Structure of Cerebrum

The functional areas of brain or cerebrum:

- (a) Frontal lobe: It consists of the motor area which controls voluntary activities. Broca's area is the motor speech area which translates thoughts into speech and controls the movements of the tongue, lips and vocal cords. The association area related to intelligence, memory, judgement and problem-solving ability is found in the frontal lobe.

- (b) **Parietal lobe:** The general sensory area associated with the sensation of temperature, touch, pressure, pain and speech is located on the post-central gyrus of the parietal lobe.
- (c) **Temporal lobe:** The area concerned with the sense of taste, sense of hearing and sense of smell are located on the temporal lobe.
- (d) **Occipital lobe:** It carries the sensory visual area concerned with the sense of vision and the association visual area for perception, analysis and storage of information obtained by sight.

